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$$\begin{aligned} & \int (e^{-3x} \cos x) dx \\ & \left\{ \begin{array}{l} u = \cos x \\ dv = e^{-3x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = (-\sin x) dx \\ v = \frac{e^{-3x}}{-3} \end{array} \right. \\ & I = \frac{e^{-3x} \cos x}{-3} - \frac{1}{3} \int e^x (-\sin x) dx \\ & \left\{ \begin{array}{l} u = \sin x \\ dv = e^{-3x} dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = (\cos x) dx \\ v = \frac{e^{-3x}}{-3} \end{array} \right. \\ & I = \frac{e^{-3x} \cos x}{-3} + \frac{e^{-3x} \sin x}{9} - \frac{1}{9} I \\ & \Rightarrow \frac{10}{9} I = \frac{e^{-3x} \cos x}{-3} + \frac{e^{-3x} \sin x}{9} + C \\ & \Rightarrow I = \frac{e^{-3x} \sin x - 3e^{-3x} \cos x}{10} + C \end{aligned}$$

## References